

Aditya Kumar Singh

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Education

Manipal Institute of Technology B.Tech in Computer Science and Engineering (Artificial Intelligence) CGPA: 8.03/10	Sep 2022 – Jul 2026 (Expected)
The Millennium School All India Senior School Certificate Examination (AISSCE), CBSE	Graduated 2021

Experience

Software Engineering Intern Eli Lilly and Company	Jan 2026 – Present
- Built and shipped features across an enterprise GenAI assistant platform (Next.js, React, TypeScript, PostgreSQL, Redis) used by 200 internal employees to create and share domain-specific AI assistants. - Integrated Veeva Vault as an AI knowledge source via MCP with SSE streaming, enabling regulated pharma document ingestion into LLM workflows; developed admin LLM governance tooling for model whitelisting and per-assistant access control. - Resolved 75% of critical/high frontend security vulnerabilities via dependency auditing and npm override strategies.	
Research Intern International Centre for Applied Sciences (ICAS)	Jun 2025 – Jul 2025
- Built a hierarchical multi-head EfficientNet-B0 classification pipeline on DERM12345 (12,345 dermoscopic images, 40 subclasses), achieving 98.32% binary accuracy and 0.994 AUC — establishing early benchmarks on a dataset published in Scientific Data (Nature, 2024) with no prior classification baselines. - Conducted a 6-model systematic study of loss functions (Dice, Tversky, Focal Tversky) on a 3D U-Net for BraTS 2020 brain tumor segmentation, achieving Dice scores of 0.77 (Tumor Core) and 0.75 (Enhancing Tumor) on 64³ voxel patches. - Deployed both models as FastAPI + React web apps with real-time inference, Grad-CAM heatmaps, and MRI segmentation overlays.	
Software Engineer Intern Recursive Zero	Jan 2025 – Feb 2025
Migrated 5+ client websites from vanilla HTML/JS to Astro.js, reducing JavaScript bundle size by 40–60% and improving Core Web Vitals and SEO performance across all projects.	

Projects

AI-Powered Running Coach Full-Stack Web App	GitHub
- Built a web-based AI running coach with GPS tracking, run history, and real-time voice feedback via Vapi, supporting 3 adaptive coaching personalities. - Designed a session analytics dashboard tracking pace, distance, and performance trends across runs.	
AI-Powered Document Intelligence Assistant	GitHub
- Built a RAG pipeline over unstructured documents (PDFs, contracts, reports) using LlamaIndex, Pinecone vector DB, and sentence-transformer embeddings for semantic search across document chunks. - Delivered real-time QA and summarization via Llama 3 (Groq) through a Next.js + FastAPI interface supporting file upload and natural language queries. - Implemented a recency-aware retrieval layer that flags conflicting or outdated information across retrieved chunks based on document timestamps, addressing knowledge staleness in production RAG systems.	

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL
Frameworks & Tools: Next.js, React, FastAPI, Node.js, Express.js, Tailwind CSS, Astro.js
AI/ML: PyTorch, TensorFlow, scikit-learn, Transformers
Databases & Infra: PostgreSQL, Redis, MongoDB, Pinecone, Docker, Kubernetes, Git